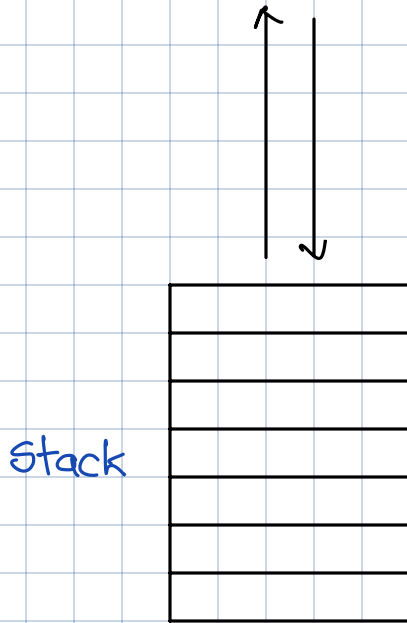
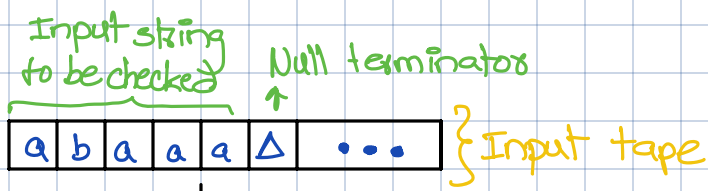




State
 START
 READ₁
 PUSH a
 READ₂
 READ₃
 POP
 READ₄
 POP
 ACCEPT

Stack
 Δ
 Δ
 aΔ
 aΔ
 aΔ
 Δ
 Δ
 Δ
 Δ

Tape
 abbbΔ
 bbΔ
 bbΔ
 bΔ
 Δ
 Δ
 Δ
 Δ

Symbols:

START

ACCEPT

REJECT

READ

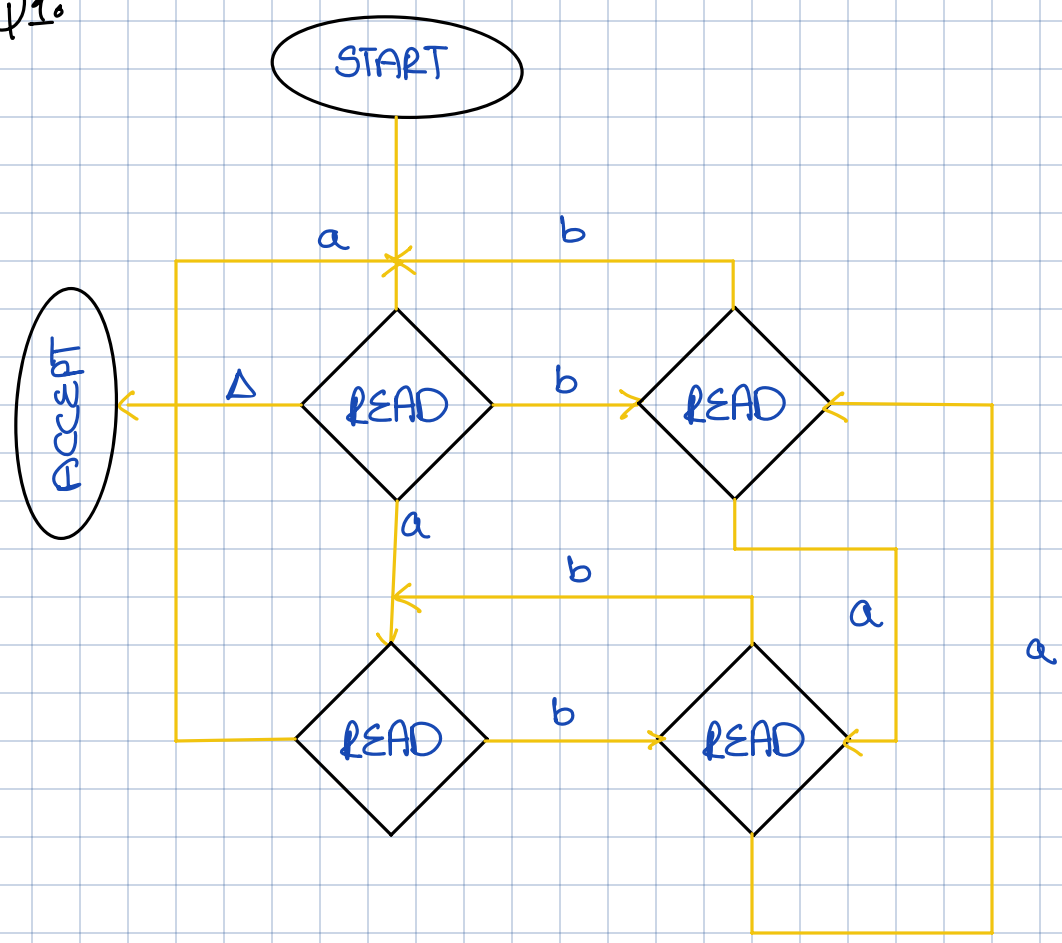
POP

PUSH

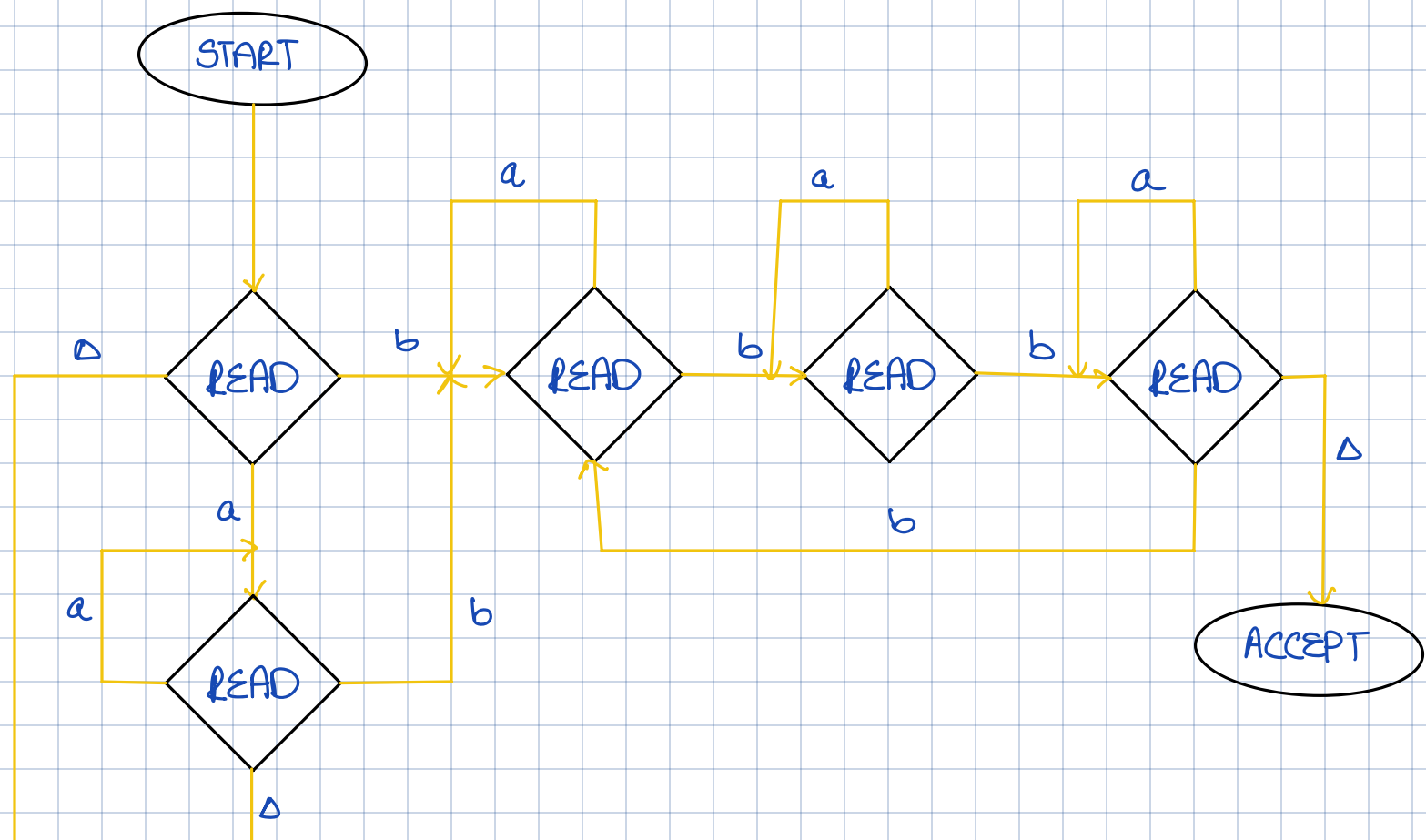
↳ It is a new kind of FA for CFs

Book Qs

Q1.



Q2.



ACCEPT

Q3.

(i) abb

State	Stack	Tape
START	Δ	abb Δ
READ ₁	Δ	bb Δ
PUSH a	a Δ	bb Δ
READ ₂	a Δ	b Δ
READ ₃	a Δ	Δ
POP	Δ	Δ
READ ₄	Δ	Δ
POP	Δ	Δ
ACCEPT		

(ii) abab

State	Stack	Tape
START	Δ	abab Δ
READ ₁	Δ	bab Δ
PUSH a	a Δ	bab Δ
READ ₂	a Δ	ab Δ
READ ₃	a Δ	b Δ
REJECT		

Q4.

(i) $a^n b^{2n}$

(ii) $S \rightarrow aSbb \mid \Lambda$

Q5.

State	Stack	Tape
START	Δ	aaabbb Δ
READ ₁	Δ	aabbb Δ
PUSH a	a Δ	aabbb Δ
READ ₂	a Δ	abbb Δ
PUSH a	aa Δ	abbb Δ
READ ₂	aa Δ	bbb Δ
PUSH a	aaa Δ	bbb Δ
READ ₂	aaa Δ	bb Δ
POP	aa Δ	bb Δ
READ ₃	aa Δ	b Δ
POP	a Δ	b Δ
READ ₃	a Δ	Δ
POP	Δ	Δ
READ ₃	Δ	Δ
POP	Δ	Δ
ACCEPT		

Q6.

(i) Yes.

(ii) $S \rightarrow aA$
 $A \rightarrow b | aAB$
 $B \rightarrow a | b$

Q8.

(i) $S \rightarrow aSb | x$
 $x \rightarrow bxa | ba$

Q10.

To accept you need to read the same no of letters as on stack. That's double. Hence even

Q11.

CFG:

$S \rightarrow XS A$
 $A \rightarrow a$
 $X \rightarrow a | b$

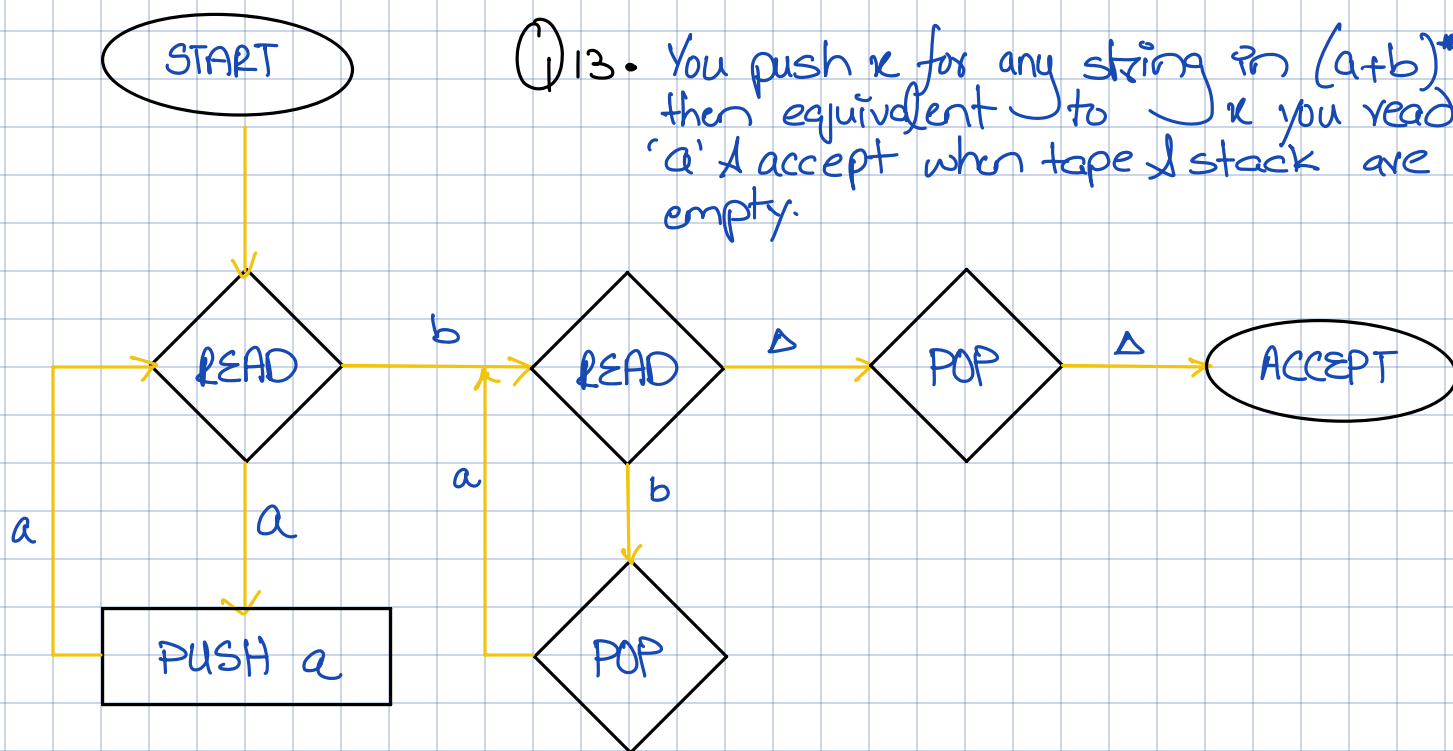
No FA possible

Q12.

State	Stack	Tape
START	Δ	$aa\Delta$
READ ₁	Δ	$a\Delta$
PUSH ₁	$x\Delta$	$a\Delta$
READ ₁	$x\Delta$	Δ
POP ₁	Δ	Δ
READ ₂	Δ	Δ
POP ₂	Δ	Δ
ACCEPT		

Q14.

Q13. You push x for any string in $(a+b)^*$ then equivalent to x you read 'a' & accept when tape & stack are empty.

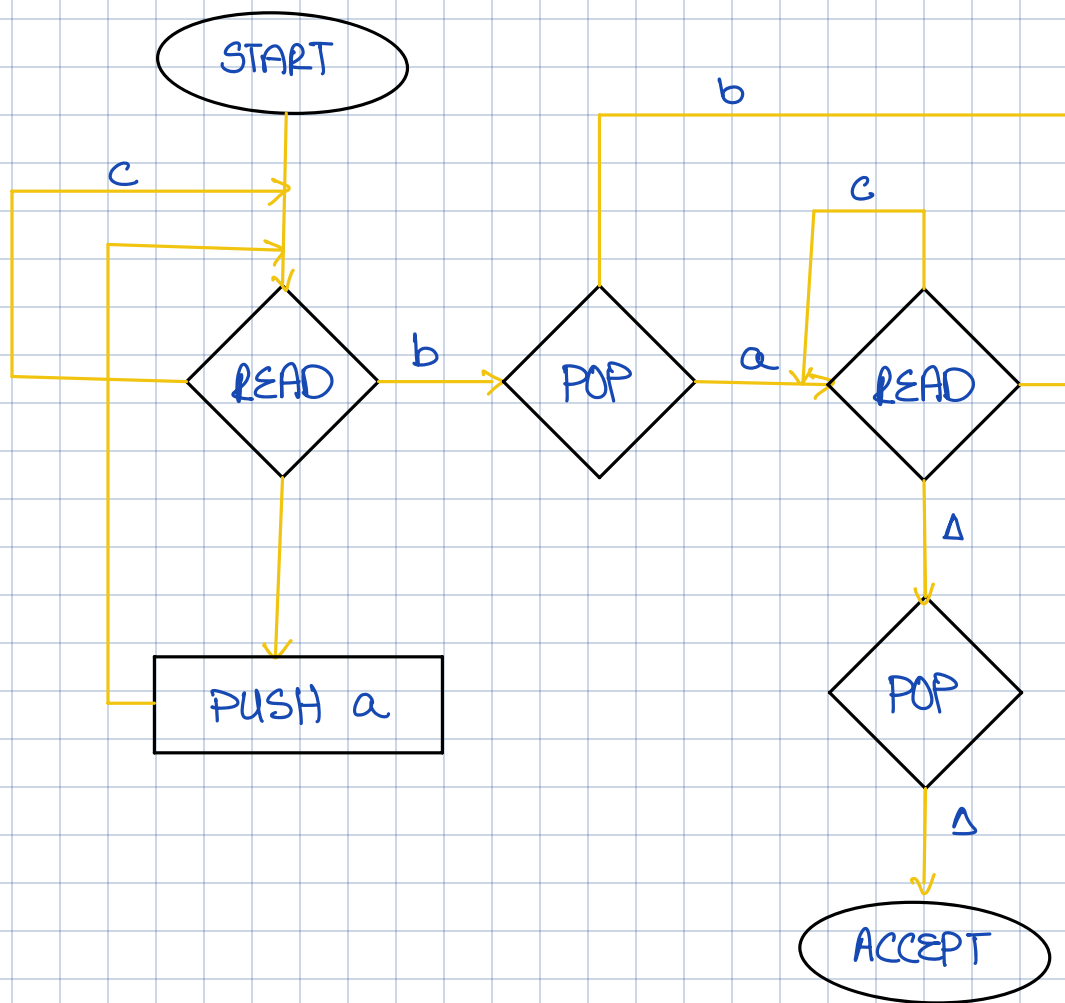


Q15.

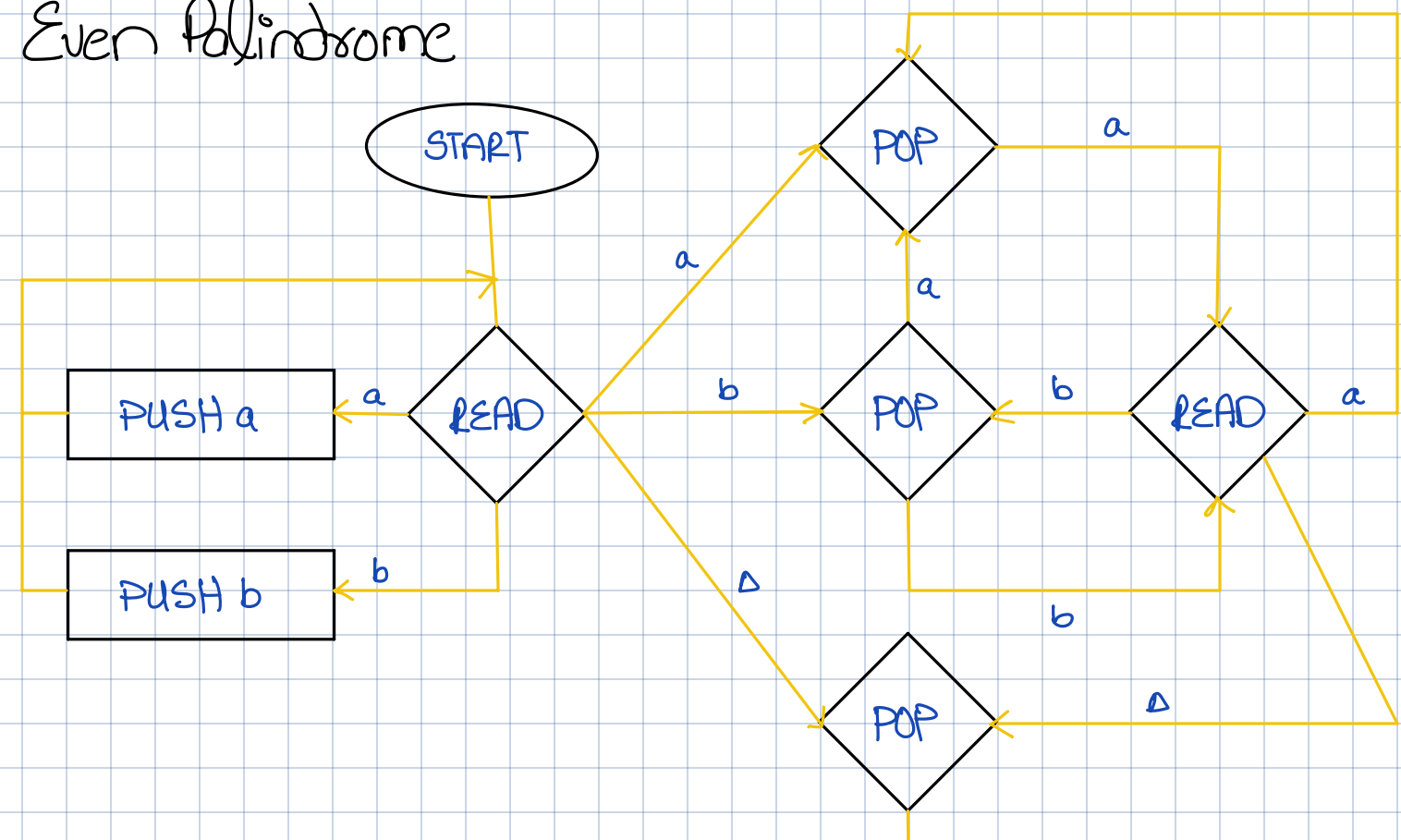
(iv) $S \rightarrow CAXBC$
 $C \rightarrow cC | \Lambda$
 $X \rightarrow AXB | C$
 $A \rightarrow aC$

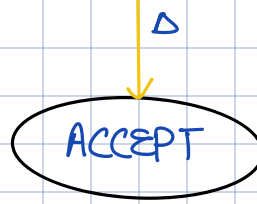
$B \rightarrow bC$

(iii)

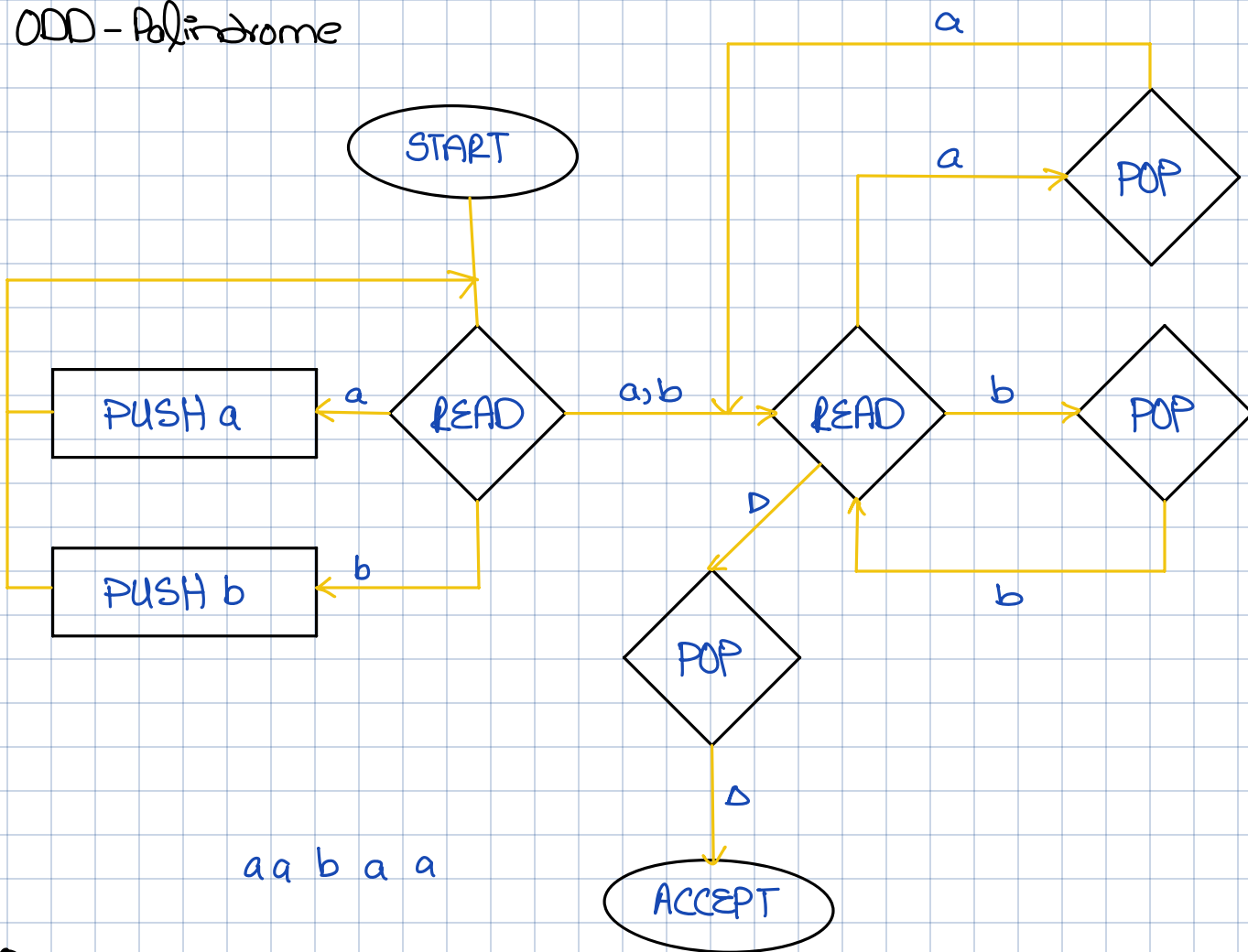


Even Palindrome





ODD - Palindrome



Q16
Palindrome:

